ESP32-WROOM-32 Wi-Fi & BLE MCU

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1. Electrical Characteristics & RF Power Specifications

ESP32-WROOM-32 is a powerful, generic Wi-Fi + Bluetooth + BLE MCU module powered by the dual-core Tensilica Xtensa 32-bit LX6 microprocessor operating up to 240 MHz.

Parameter	Specification	Engineering Note
Operating Voltage (VDD)	3.0V to 3.6V DC	STRICT: Applying 5.0V destroys the chip!
Peak Power Supply Current	500 mA minimum capability	Wi-Fi TX bursts draw up to 450 mA
Digital I/O Voltage	3.3V LVTTL logic	Inputs are NOT 5V tolerant
Deep-Sleep Current	5 uA to 15 uA	Optimal for ultra-low power IoT sensors
Operating Frequency	Up to 240 MHz (600 DMIPS)	Internal 520 KB SRAM + 4 MB Flash

2. Critical Strapping Pins & Boot Modes

The ESP32 module has 5 strapping pins that configure the chip state at power-up / reset. Incorrect external pullups will cause boot failures:

• GPIO0: Boot Mode Selection. During reset, pull HIGH for normal SPI Flash boot. Pull LOW to enter UART Download/Flashing mode.

• GPIO2: Flashing constraint. Must be left floating or pulled LOW during flashing. Connected to onboard LED on many dev boards.

• GPIO12 (MTDI): Flash Voltage Selection. Must be LOW at boot to configure internal flash LDO to 3.3V. If pulled HIGH, flash is overvolted to 1.8V and chip panics.

• GPIO15 (MTDO): Debug log output. Must be pulled HIGH to enable silent boot; LOW outputs debug ROM messages.

• EN (CHIP_PU): Enable pin. Must have 10 kOhm pullup to 3V3 and 1 uF capacitor to GND to delay boot until power rail stabilizes.

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3. Power Supply Design & Decoupling Recommendations

Wi-Fi Transmission Demands: When the 2.4 GHz RF power amplifier activates, current jumps from 30 mA to 450 mA within 10 nanoseconds. Standard USB-to-UART bridge LDOs (e.g. AMS1117-3.3) fail if decoupling is inadequate.

Mandatory Decoupling Architecture:

• 1x 10 uF low-ESR ceramic or tantalum capacitor located immediately adjacent to the VDD pin.

• 2x 0.1 uF ceramic capacitors in parallel for high-frequency attenuation.

• Total power rail impedance must be less than 0.2 Ohms to prevent voltage sag below 2.8V.

4. Troubleshooting Diagnostic Guide

• Error: 'Brownout detector was triggered at core 0 / 1'. Cause: The 3.3V power supply collapsed below 2.8V during Wi-Fi connection handshake. Solution: Use dedicated 3.3V regulator rated for >= 800 mA (e.g., AP2112K or LM1117-3.3) with large 470 uF bulk capacitor.

• Error: 'A fatal error occurred: Failed to connect to ESP32: Timed out waiting for packet header'. Cause: GPIO0 is not held LOW during bootloader synchronization, or EN capacitor is missing. Solution: Hold BOOT button (pulls GPIO0 low) while pressing and releasing EN (Reset).

• Error: GPIO input reading false positives or ghost triggers. Cause: Floating inputs. ESP32 internal pullups are relatively weak (~50 kOhm). Solution: Add external 10 kOhm pullup or pulldown resistors on sensor lines.